IBM Project 1 - Validation and AI-use Record

Prepared: October 9, 2026, America/Chicago
Valuation cutoff: September 1, 2026 after close

Actual reproducible runs

`python3 -m unittest discover -s tests -v` completed with **24 passed, 0 failed**. Tests cover linked statements, the H1 actual plus H2 forecast boundary, cash, FCFF, terminal economics, valuation bridge, sensitivity monotonicity, peer arithmetic and validation ranges.

`python3 model/ibm_phase3_model.py --write-output --demo-failure` regenerated the model output and detected the intentional $1m FY2027 cash break. The break occurs in a copy and leaves the saved base case unchanged. `python3 app/server.py --write-static` regenerated the visible base case. The cold-run record reports a local HTTP 200 API response and the same $181.950342 per-share value.

Financial-source checks

The IBM Q2 2026 Form 10-Q anchors the June 30 balance sheet and H1 results. Two IBM investor-relations dividend records agree on the $1.69 dividend, August 10 record date, and September 10 payment. It was known but unpaid on the September 1 valuation date, so the bridge deducts it once as a payable. The source ledger contains URLs, definitions, reporting dates and cutoff availability.

Failures, limits and recovery

The project preserved a prior failure in which full-year 2026 cash/equity flows were improperly added to the June opening balance sheet. The repaired model rolls only H2 forecast activity. The current deliberate failure reduces FY2027 cash by $1m and correctly fails the balance-sheet check; the next base run restores the valid output. No course reference answers were available, so no actual known-answer test is claimed.

> **Unresolved evidence:** opaque beta methodology, generic rather than IBM-specific debt-yield proxy, weighted-average diluted shares as a point-in-time proxy, analyst cash reserve, zero incremental pension claim, and book-value Financing sidecar. These constraints support Watch / Defer rather than an initiation.

AI-use disclosure

On October 9, 2026, an OpenAI Codex agent helped inspect code, run tests, repair the P/E definition, reconcile outputs, and format HTML/PDF artifacts. It did not access course answers, submit the project, trade securities, or claim a proprietary data source. The environment did not provide model/version metadata, so none is asserted. The analyst remains responsible for all judgment and source verification.